

IFQ714 Introduction to JavaScript Programming. Assignment 1: Programming exercises
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21Aug26
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Introduction

Near Earth Objects (NEOs) are asteroids or comets which have an orbit path bringing them within 1.3 AUs of the sun, and approximately 45 million km from the Earth's orbit path.

The goal of this assignment is to perform an analysis on the provided NEO dataset, extract some valuable insights, and return the dataset to JSON format re-ordered by orbit class.

Data Format

NASA data was provided in JSON format. In order to read the file:

1. A new JavaScript (JS) file was created within same folder location.
2. The JavaScript file was then 'linked' to the JSON (exported non-global), and formatted (JSON.parse) to enable further analysis.

Functions

To gather data insights, the following basic functions were utilised:

- filter()
- sort()
- slice()
- map()
- toFixed()

Data Analysis

Before commencing analysis, I wanted to ensure that all JSON data had been correctly parsed, using the .length property.

Q: how many Apollo class NEOs are considered potentially hazardous asteroids?

A constructor was created using the filter array method. A strict equality operator was applied against the orbit_class data, with logical 'and' operator to assess .pha value 'true'.

Q: what is the average perihelion for the Apollo NEO class?

Orbit distance closest to the sun was provided within the q_au_1 data column. Average perihelion was calculated by adding all numbers, and dividing by total count; with .toFixed() method utilised to round decimal digits (2).

Q: can you please identify and list the ten largest NEOs?

For this data request, I sorted the array using the arrow function. After the array had been sorted (h_mag), I used the slice method to extract the required section.

Reformatting data

After sorting data by class, it is necessary to re-format the dataset to universal JSON plain-text. This task is achieved by using the built-in `JSON.stringify()` method. The new file is named and created by executing `fs.writeFileSync`.

Unit test

Apollo orbit class filter and math function output was verified using the Jest testing framework. The test confirmed expected results for 'largest 10 NEO' and Apollo object sort by quantity. 'not.toBe' check was used to validate non-shared memory reference for the deep copy.

Challenges / issues

I had challenges with 'calling out' parsed data into required functions, often receiving errors that I had not idea how to rectify. Further investigation identified syntactical anomalies, and this was a great learning experience. I had similar issues when importing data for jest evaluation (I had issues with non-conventional file name (must be 'xx.test.js')).

Only through (1)many hours of research, (2)combining learning material with 'trial and error', (3)and desperate requests for OLA assistance, have I managed to accomplish this assignment task.

Incomplete tasks

I have not achieved my goal of extracting max / min / average data across all column values for the Aten orbit class (qty 15 Aten NEOs).

One of my initial goals was to extract an array of objects directly into working .js > identify all Apollo class NEOs with the same perihelion > map their aphelion > and predict next perihelion date (from today) by calculating orbital period - I just can't confirm at what stage of orbit the NEO was discovered (I may find this piece of data whilst continuing to work on this code).

Conclusion

Having no coding experience, this assignment has been a significant challenge (I didn't even know what JavaScript was...).

Learning how to setup and work with new software, as well as understanding structures and machinations of the JavaScript language -it has been an exciting journey so-far.

I now have a solid basic understanding of data management 'behind the scenes', and am looking forward to further foundational studies to support my learning of the JavaScript engine.

References:

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